

$$a = 215.54279 \in \mathbb{R}$$

$$N = 10$$

$$0.21554279 \cdot 10^9$$

p

$$a = p \cdot N^q$$

p

$$p \in \mathbb{R} \quad q \in \mathbb{Z}$$

p = mantissa
q = caratteristica

$$0.1 \leq |p| < 1 \quad \text{Forma normalizzata}$$

$$p = \pm 0.p_1 p_2 p_3 \dots \text{ con } p_1 \neq 0$$

$$0.021554279 \cdot 10^4 \rightarrow \text{NON NORMALIZZATA } p < 0.1$$

$$2.1554279 \cdot 10^2 \rightarrow \text{NON NORMALIZZATA } p > 1$$

IN BASE 10

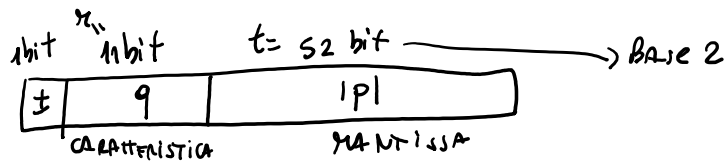
$$215.54279 = 5 \cdot 10^0 + 1 \cdot 10^1 + 2 \cdot 10^2 + 5 \cdot 10^{-1} + 4 \cdot 10^{-2} + 2 \cdot 10^{-3} + 7 \cdot 10^{-4} + 9 \cdot 10^{-5}$$

IN BASE 2

$$1061.1101 \rightarrow 1 \cdot 2^0 + 0 \cdot 2^1 + 0 \cdot 2^2 + 0 \cdot 2^3 + 1 \cdot 2^{-1} + 1 \cdot 2^{-2} + 0 \cdot 2^{-3} + 1 \cdot 2^{-4}$$

$$a = p \cdot N^q$$

$$0.1 \leq |p| < 1 \quad \text{Forma normalizzata}$$



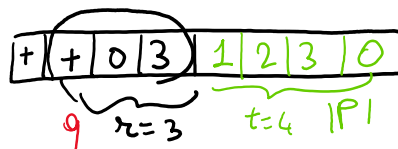
L più piccolo intero memorizzabile

U più grande intero memorizzabile

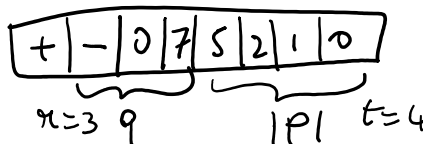
$$L \leq q \leq U \quad \underline{\text{OK}}$$

$$N = 10 \quad t = 4 \text{ cifre} \quad \pi = 3 \text{ cifre}$$

$$a = 123 = 0.123 \cdot 10^3$$



$$a = 0.521 \cdot 10^{-7}$$



$$r=3 \quad q^- \quad |p| \quad t=4$$

$$N=10, t=4, \underline{r=3}$$

$$a = 0.215 \cdot 10^{350} \quad \text{overflow}$$

$$a = -0.324 \cdot 10^{-527} =$$

$$= -0. \underbrace{000000 \dots 000}_{527 \text{ zeri}} 324$$

$$a = p \cdot 10^q$$

$$q > U$$

IL NUMERO NON
PUO' ESSERE
MEMORIZZATO

OVERFLOW

→ MATLAB/PYTHON → inf
In doppia precisione $U \approx 308$

$$q < L$$

il non può essere
memorizzato

UNDERFLOW

IN MATLAB/PYTHON

$$a = 0$$

IL NUMERO VIENE MEMORIZZATO
COME ZERO



Command Window

New to MATLAB? See resources for [Getting Started](#).

>> realmax

ans =

1.7977e+308

>> a=2e308

a =

Inf

>> a=1.5e308

a =

1.5000e+308

Command Window

New to MATLAB? See resources for [Getting Started](#).

>> realmin

ans =

2.2251e-308

>> a=5.6e-350

a =

0

>> a=6.7e-311

a =

6.7000e-311

è il più piccolo numero memorizzabile nello
standard IEEE
(forma normalizzata)

per capire
il perché

Esempio

$$N=10$$

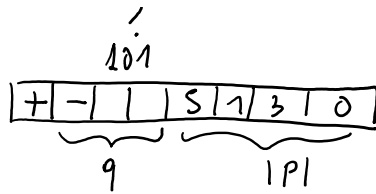
$$r=3$$

$$a = 0.513 \cdot 10^{-101}$$

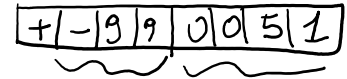
$$L = -99$$

$$\frac{0.00513}{10^1} \cdot 10^{-99}$$

$$0.00513 \cdot 10^{-99}$$



$$0.00513 \cdot 10^{-23}$$



FORMA
DENORMALIZZATA

Esponento più
piccolo che
risale
a memoria $L-t$

il più piccolo numero memorizzabile in forma
de-normalizzata è dell'ordine di $\sim 10^{-324}$

```
>> a=4.7e-324
```

```
a =
```

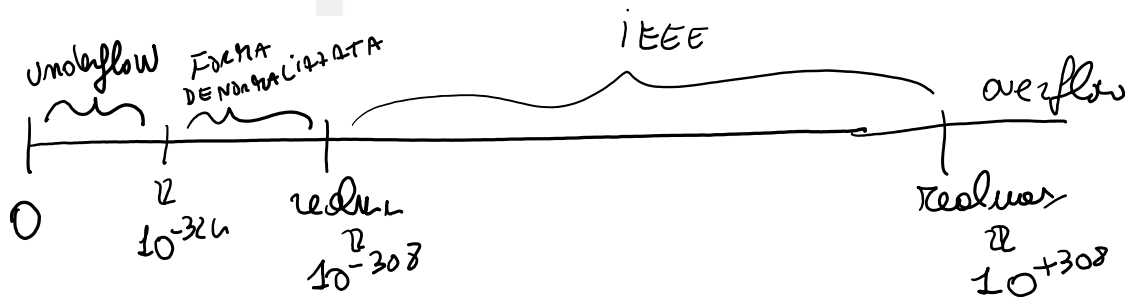
```
4.9407e-324
```

```
>> a=5.8e-325
```

```
a =
```

```
0
```

1a)



$$a = p \cdot N^q$$

$$\frac{1}{N} \leq |p| < 1$$

$$L \leq q \leq U$$

CARATTERISTICA

t numero di cifre a disposizione per |p|

mantissa

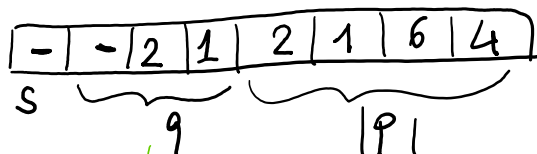
ESEMPIO

$$N=10$$

$$t=3$$

$$L=4$$

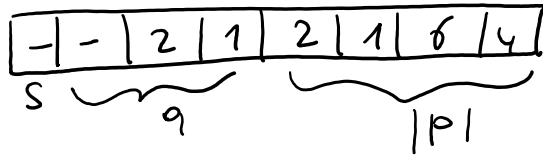
$$a = -0.2164 \underline{3} 12 \cdot 10^{-21}$$



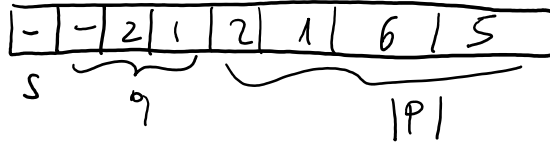
TROVARE
ARROTONDARE

$$a = -0.2164 \underline{5} 12 \cdot 10^{-21}$$

$$a = -0.2164912 \cdot 10^{-21}$$



TRONCAMENTO



ARROTONDAMENTO

$p > 0$

$$a = p \cdot N^q$$

$$p = 0.p_1 p_2 p_3 \dots p_k p_{k+1} p_{k+2} \dots$$

SE $p_{k+1} = p_{k+2} = \dots = 0$
 a è un numero macchina

Se a non è un numero macchina allora p deve essere accorciato

$\swarrow$ per TRONCAMENTO
 $\searrow$ per ARROTONDAMENTO

$\bar{a}$ numero memorizzato su calcolatore $\bar{a} \approx a$

$$\bar{a} = \bar{p} \cdot N^q$$

TRONCAMENTO

$$\bar{p} = 0.p_1 p_2 \dots p_k$$

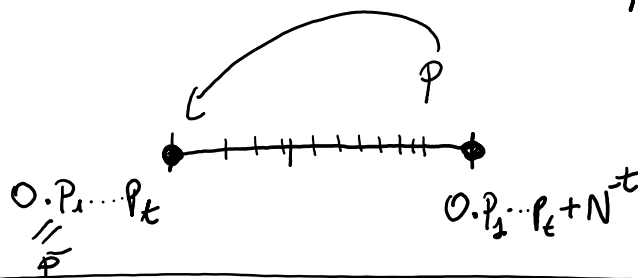
ARROTONDAMENTO

$$\bar{p} = \begin{cases} 0.p_1 \dots p_k & \text{se } p_{k+1} < \frac{N}{2} \\ 0.p_1 \dots p_k + \underbrace{0.0 \dots 01}_{N^{-t}} & \text{se } p_{k+1} \geq \frac{N}{2} \end{cases}$$

$$p = 0.p_1 \dots p_k p_{k+1} \dots$$

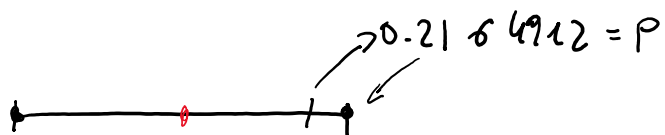
TRONCAMENTO

$$|p - \bar{p}| < N^{-t}$$

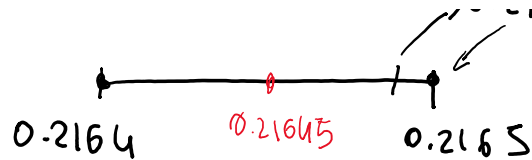


$$p = 0.2164912$$

$$t = 4$$

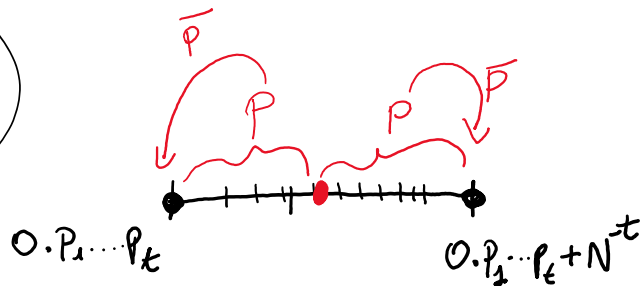


$t=4$



ARROTONDAMENTO

$$|p - \bar{p}| \leq \frac{N}{2} \cdot 10^{-t}$$



$w = p \cdot N^q \longrightarrow \bar{a} = \bar{p} \cdot N^q$ ARROTONDAMENTO

$$\epsilon_R = \frac{|w - \bar{a}|}{|w|} \leq \frac{1}{2} \cdot N^{1-t}$$

SI DIMOSTRA (LA PROSSIMA VOLTA)

PRECISIONE DI MACCHINA ϵ_{PS}

MATLAB

```
Command Window
New to MATLAB? See resources for Getting Started.

>> (1/2)*2^(1-52)

ans =

    2.2204e-16

>> eps

ans =

    2.2204e-16

fx >> |
```